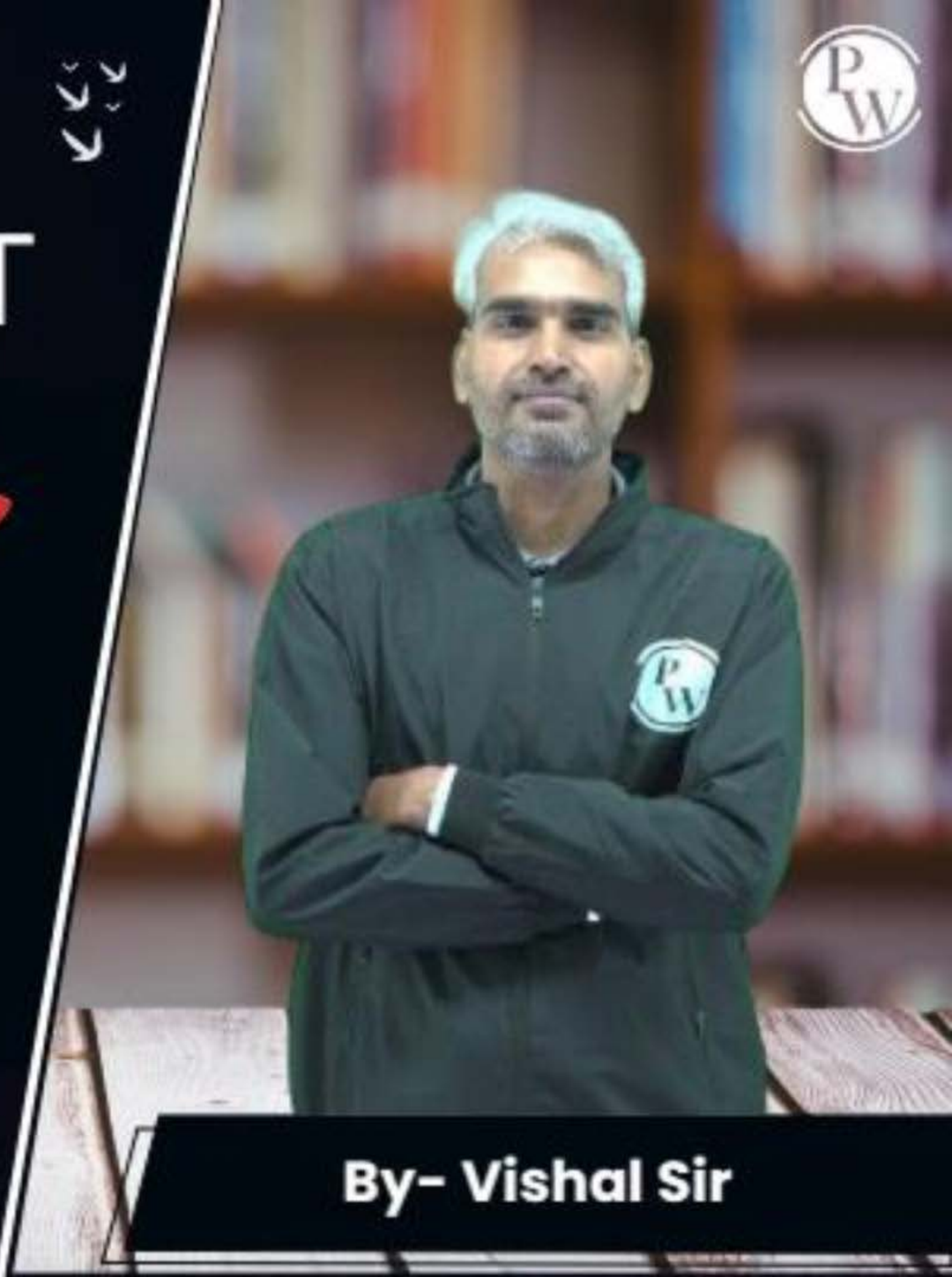
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 06



By- Vishal Sir

Recap of Previous Lecture



Topic — Properties of set operations

Topic — Multi-set

Slide

Topics to be Covered



Topic Multi-set

Topic Cartesian product

Topic Relation and types of relations

Slide



Topic : Multi-set

- An unordered Collection of objects/elements in which an element may appear more than once is Called a multiset.

eg. Multiset = $\{a, b, b, a, a, c, d, d\}$

(or frequency) = $\{3.a, 2.b, 1.c, 2.d\}$

These numbers represent the multiplicity of Corresponding element

Slide

Let multiset $P = \{ m_1.a_1, m_2.a_2, m_3 a_3, \ldots\ldots, m_k.a_k \}$

where 'm_i' is the multiplicity of element a_i.

And

let multiset $Q = \{ n_1.a_1, n_2.a_2, n_3.a_3, \ldots\ldots, n_k.a_k \}$

where 'n_i' is the multiplicity of element a_i.

Find multiplicity of a_i in

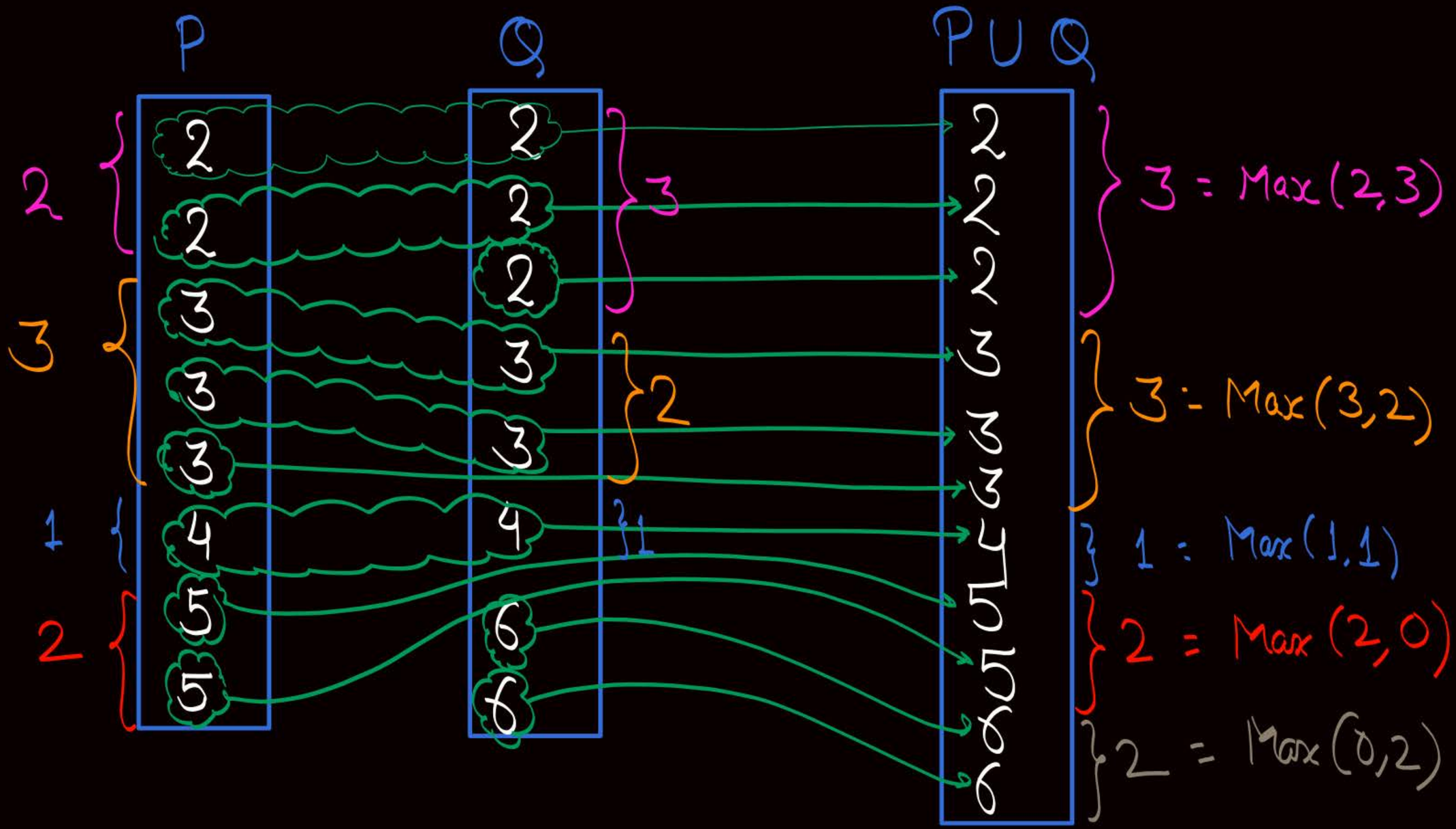
Q.t multi-set

(i) $P \cup Q$

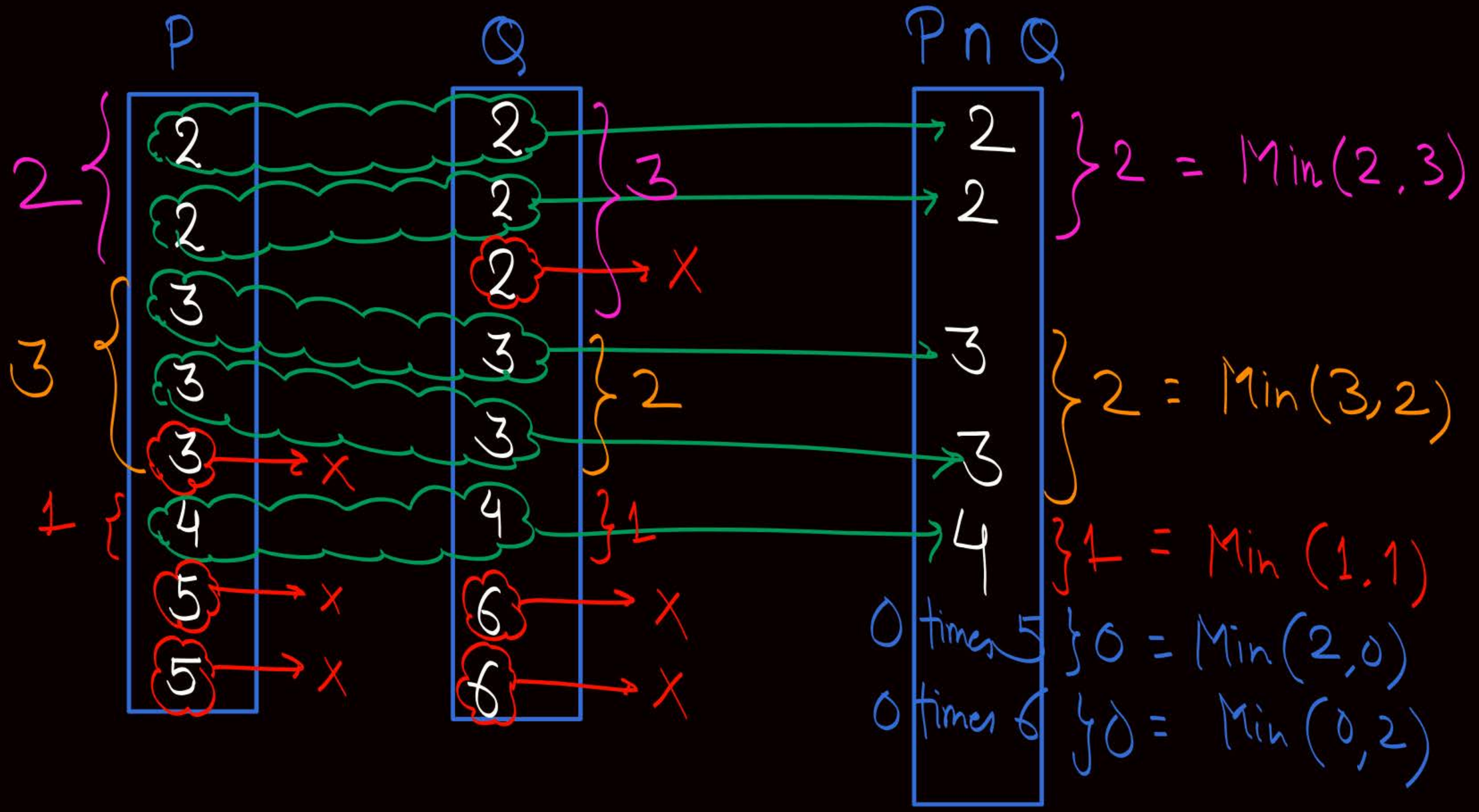
(ii) $P \cap Q$

(iii) $P - Q$

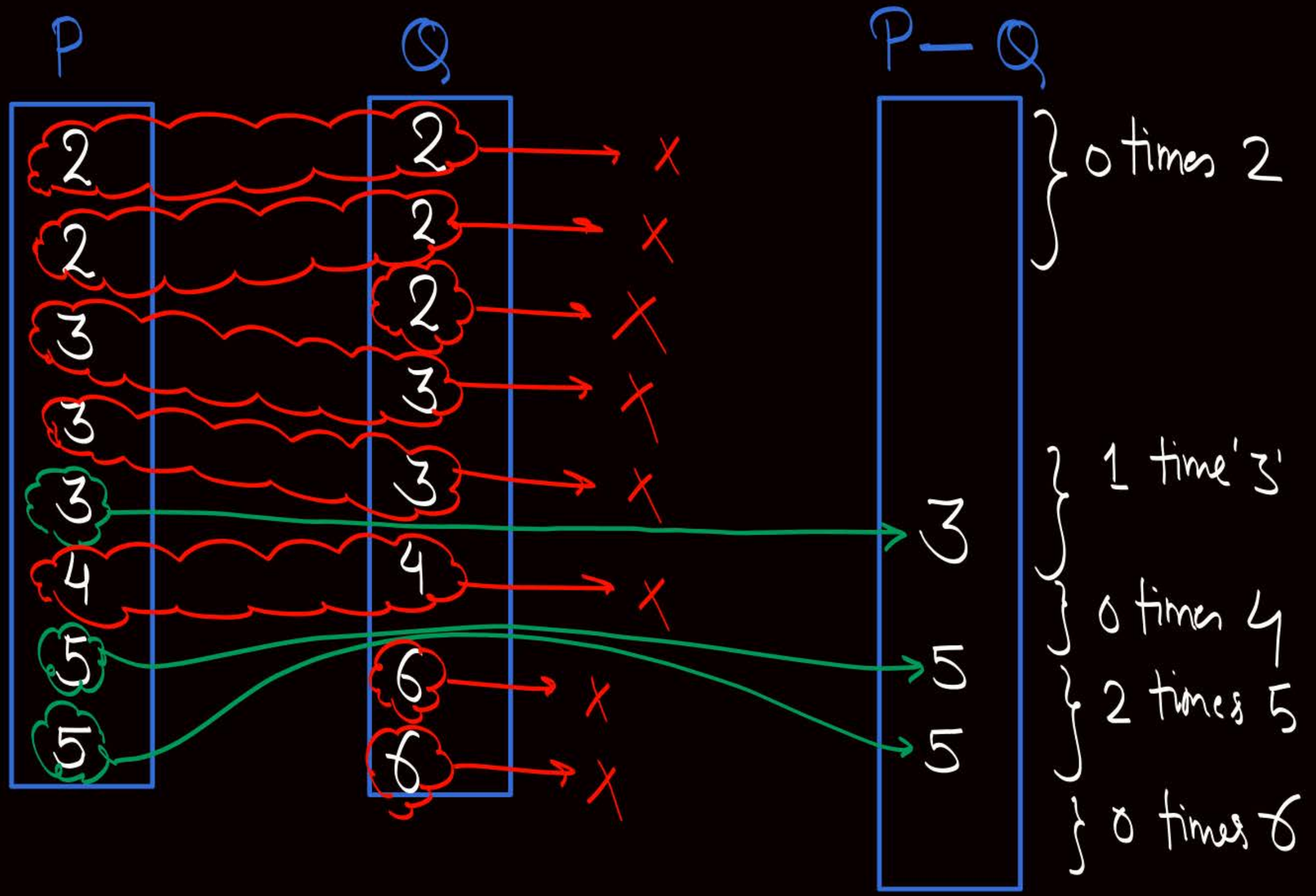
Multiset P ∪ Multiset Q



Multiset P ∩ Multiset Q



Multiset P — Multiset Q



Let multiset $P = \{ m_1.a_1, m_2.a_2, m_3 a_3, \ldots\ldots, m_k.a_k \}$

where 'm_i' is the multiplicity of element a_i.

And

let multiset $Q = \{ n_1.a_1, n_2.a_2, n_3.a_3, \ldots\ldots, n_k.a_k \}$

where 'n_i' is the multiplicity of element a_i.

Find multiplicity of a_i in

Output multi-set

(i) $P \cup Q$ = Multiplicity of a_i in $P \cup Q$ = $\max(m_i, n_i)$

(ii) $P \cap Q$ = Multiplicity of a_i in $P \cap Q$ = $\min(m_i, n_i)$

(iii) $P - Q$ = Multiplicity of a_i in $P - Q$ = $\begin{cases} m_i - n_i, & \text{if } m_i > n_i \\ 0, & \text{otherwise} \end{cases}$

Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many different multisets can be created using the element of set A.

{ Any element of set A can be chosen any number of times }

Multisets possible with element '1' of set A.

$M_1 = \{1\}$ Size of $M_1 = 1$

$M_2 = \{1, 1\}$ Size of $M_2 = 2$

$M_3 = \{1, 1, 1\}$ Size of $M_3 = 3$

$M_{100} = \{\overbrace{1, 1, 1, \ldots 1}^{100 \text{ times}}\}$ Size of $M_{100} = 100$

$M_\infty = \{1, 1, \ldots$ infinite times $1\}$ Size of M_∞ = not definite

If we do not have any restriction on the size of multi-set, then infinite multi-sets are possible with a single element.

∴ Ans = infinite

Today's Topic

H.W. Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many different multisets of 'size = 4' can be formulated using the elements of set A, such that at least one element of set A appears exactly twice in each such multiset.

Summation of multiplicities = 4

$A = \{1, 2, 3, \ldots\ldots n\}$

Based on given constraint only two types of multi-sets are possible

✓ Type ② $\{a, a, b, c\}$ or ✓ Type ② $\{a, a, b, b\}$

① $\left({}^{n}C_1 * 1\right) * \left(\frac{{}^{(n-1)}C_1 * {}^{(n-2)}C_1}{2}\right) = \boxed{\frac{n(n-1)(n-2)}{2}}$

$\frac{{}^{n}C_1 * 1 * {}^{(n-1)}C_1 * 1}{2} = \boxed{\frac{n(n-1)}{2}}$

② $\left({}^{n}C_1 * 1\right) * {}^{(n-1)}C_2 * 1 = \boxed{n \cdot \frac{(n-1)(n-2)}{2}}$

$\left({}^{n}C_2 * 1\right) = \boxed{\frac{n(n-1)}{2}}$

③ $\left({}^{n}C_3\right) * {}^{3}C_1 * 1 * 1 = \frac{n(n-1)(n-2)}{\cancel{3} \cdot 2 \cdot 1} * \cancel{3} \times 1 \times 1$

$= \boxed{\frac{n(n-1)(n-2)}{2}}$

H.W. Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many different multisets of size = 4 can be formulated using the elements of set A, such that at least one element of set A appears exactly twice in each such multiset.

Summation of multiplicities = 4

The required multisets can be of type-① or type-② only

∴ Total no. of such multisets possible:

= Using type ① + Using type-2

$$= \frac{n(n-1)(n-2)}{2} + \frac{n(n-1)}{2} = \boxed{\frac{n(n-1)^2}{2}}$$

Ans

HoHo

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots n\}$

How many different multisets of 'size = 4' can be formulated using the elements of set A.

(i) $\{a, b, c, d\} = {}^{n}C_4 = \frac{n(n-1)(n-2)(n-3)}{24}$

(ii) $\{a, a, b, c\} = {}^{n}C_3 * {}^{3}C_1 * 1 = \frac{n(n-1)(n-2)}{2}$

(iii) $\{a, a, b, b\} = {}^{n}C_2 = \frac{n(n-1)}{2}$

(iv) $\{a, a, a, b\} = {}^{n}C_2 * {}^{2}C_1 = n(n-1)$

(v) $\{a, a, a, a\} = {}^{n}C_1 = n$

Total no of multisets Possible

$$= \left(\frac{n(n-1)(n-2)(n-3)}{24}\right) + \left(\frac{n(n-1)(n-2)}{2}\right) + \left(\frac{n(n-1)}{2}\right) + \left(n(n-1)\right) + n$$

Relation

Cartesian Product / Cross Product

Let A & B are two sets.
Cartesian product of A and B is denoted by 'A x B' and it is defined as,

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

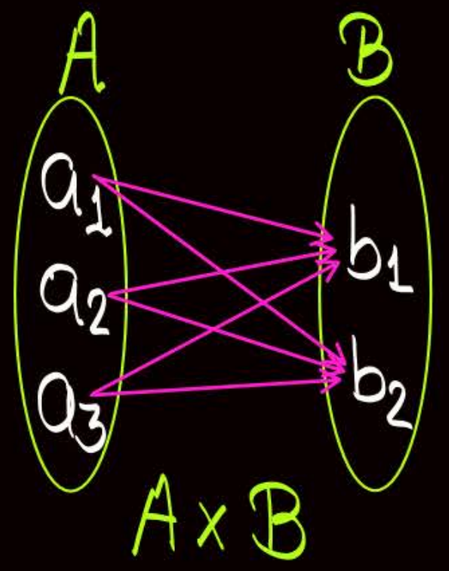
Order pair
i.e., order is important
$(a, b) \neq (b, a)$

first element will be from 1st set

2nd element of order pair will be from 2nd set.

eg: Let $A = \{a_1, a_2, a_3\}$

$B = \{b_1, b_2\}$



$A \times B = \{(a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2)\}$

$B \times A = \{(b_1, a_1), (b_1, a_2), (b_1, a_3), (b_2, a_1), (b_2, a_2), (b_2, a_3)\}$

We know, $(a_1, b_1) \neq (b_1, a_1)$ $\therefore$ in general $A \times B \neq B \times A$

In general,

$$A \times B \neq B \times A$$

- "$A \times B = B \times A$" if and only if

  (i) Either $A = B$

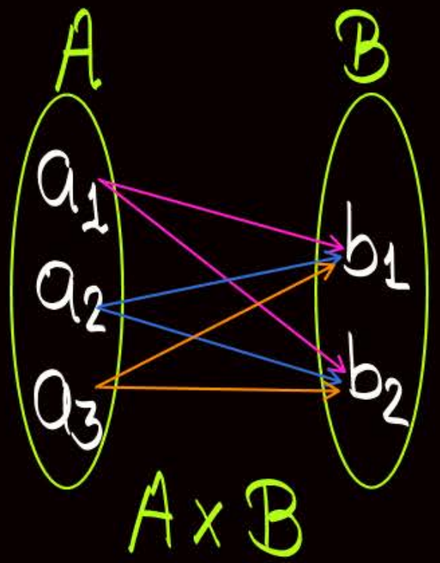
  Or (ii) At least one set out of A or B is an Empty set.

If at least one of A or B is an Empty set,
then $A \times B = B \times A : \{\ \}$

Result will be an Empty set

eg: Let $A = \{a_1, a_2, a_3\}$

$B = \{b_1, b_2\}$



$A \times B = \{(a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2)\}$

Note:- We can observe that in $A \times B$ every element of set A relates with every element of set B.

∴ If $|A| = m$ & $|B| = n$

then $|A \times B| = m.n. = |A| \cdot |B|$



Topic : Relation

* A relation from set A to set B defines that how exactly the elements of set A relates with the elements of set B.

* In Cartesian product 'A × B' Every element of set A relates with every element of set B.
  i.e 'A × B' is the universal set w.r.t. relations from set A to set B.

> Any relation from set A to set B is a subset of A × B

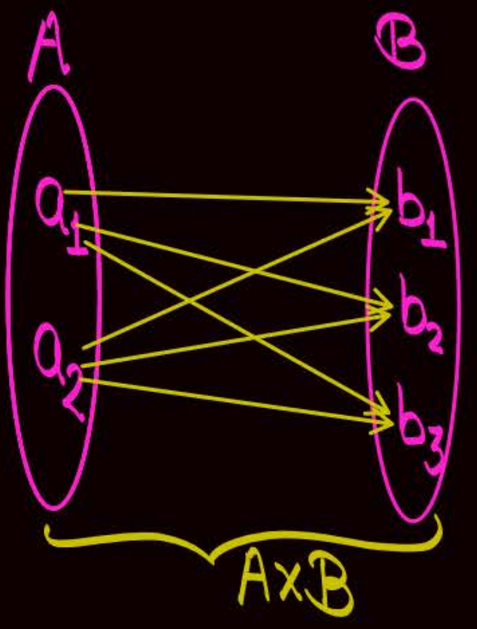
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eg:-

$A = \{a_1, a_2\}$

$B = \{b_1, b_2, b_3\}$

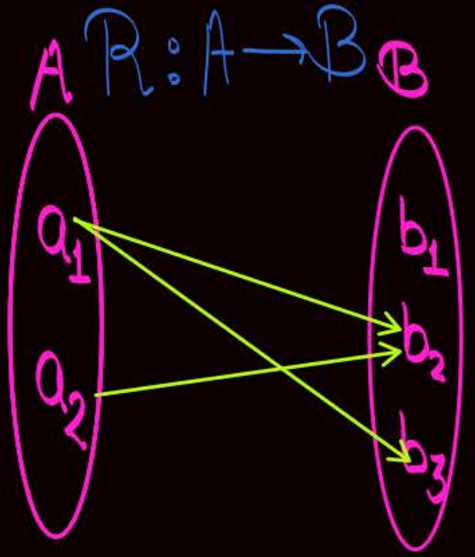
* $A \times B = \{(a_1, b_1), (a_1, b_2), (a_1, b_3), (a_2, b_1), (a_2, b_2), (a_2, b_3)\}$



+ Let 'R' denotes a relation from set A to set B defined as

$R = \{(a_1, b_2), (a_1, b_3), (a_2, b_2)\}$

We can observe that all order pairs of relation R are member of $A \times B$ ie R is a subset of $A \times B$



* Number of different relations possible from set A to set B

- Let A and B are two sets such that $|A| = m$ and $|B| = n$, then

* How many different relations are possible from set A to set B?

- Every relation from set A to set B is a subset of $A \times B$

∴ Total no. of relations possible from set A to set B = Number of subsets of $A \times B$

$$= 2^{|A \times B|} = 2^{|A| \cdot |B|}$$

$$= 2^{m \cdot n}$$

- One subset of 'A x B' is Empty set.
- This Empty subset of A x B is also a relation from set A to set B; and it is called "Empty relation" from set A to set B.



Topic : Relation

☆ A relation from set A to set A it self is called a relation on set A

* Every relation on set A is a subset of "A × A"

- If $|A| = n$, then

Number of relations possible on set A = Number of subsets of A×A

$= 2^{|A \times A|} = 2^{|A| \cdot |A|}$

$= 2^{n \cdot n} = 2^{(n^2)}$

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Topic : Types of Relations

All these relations are defined on same set {i.e from set A to set A itself}

(1) Diagonal Relation (Identity Relation)

(2) Reflexive Relation

(3) Irreflexive Relation

(4) Symmetric Relation

(5) Anti-symmetric Relation

(6) Asymmetric Relation

(7) Transitive Relation

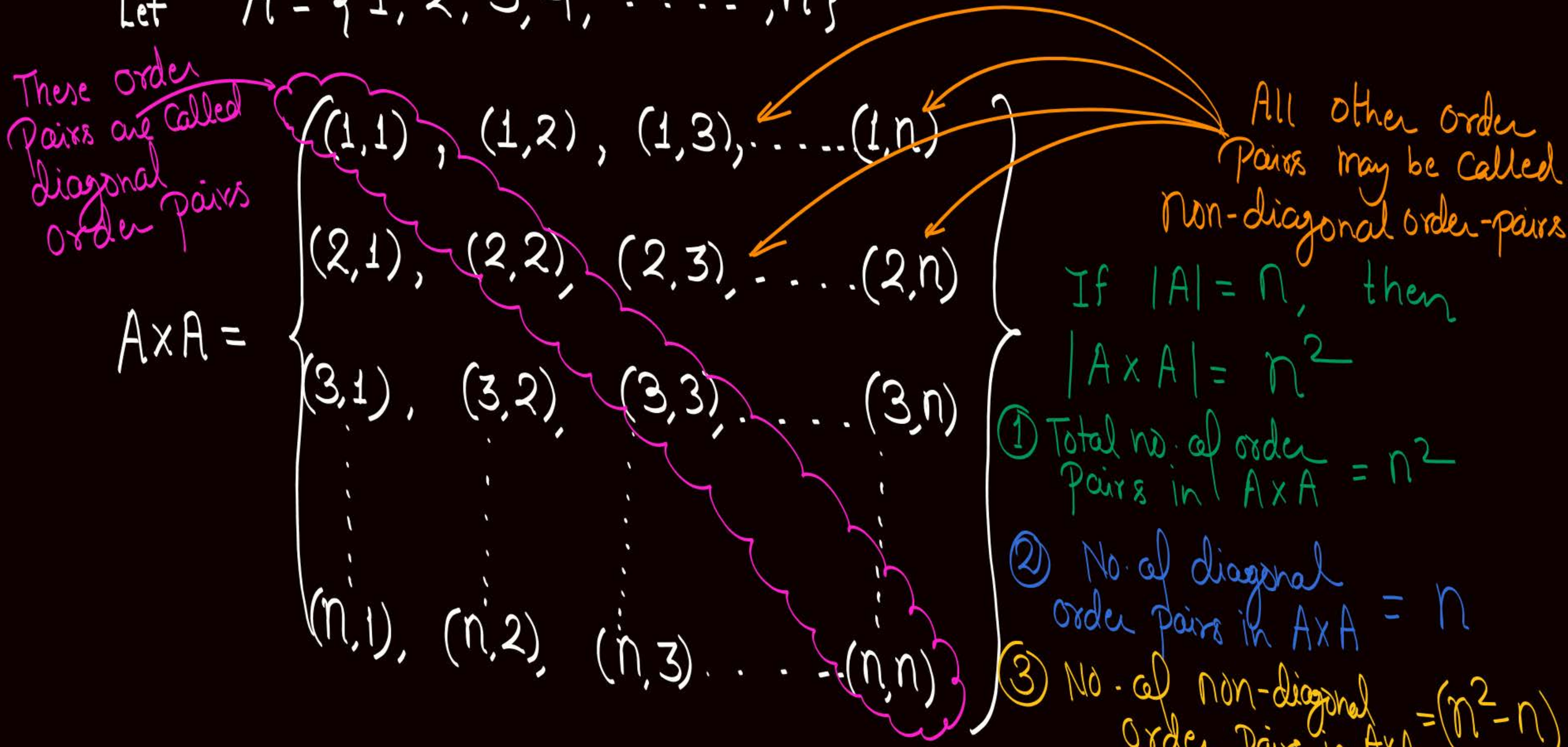
(8) Complement of a relation

(9) Inverse of a relation

(10) Composite of two relations

Slide

Let $A = \{1, 2, 3, 4, \ldots, n\}$



If $|A| = n$, then

$|A \times A| = n^2$

① Total no. of order pairs in $A \times A = n^2$

② No. of diagonal order pairs in $A \times A = n$

③ No. of non-diagonal order pairs in $A \times A = (n^2 - n)$



Topic : Diagonal Relation (Identity Relation)

- A diagonal relation on set A (i.e. identity relation on set A) can be denoted by Δ_A (or I_A) and it is defined as,

$$\Delta_A = \{(a,a) \mid a \in A\}$$ ← it is the definition of the set.

eg. Let $A = \{1, 2, 3\}$

then $\Delta_A = \{(1,1), (2,2), (3,3)\}$ it is diagonal relation on set A.

$R_1 = \{(1,1), (2,2)\}$ $3 \in A$, but $(3,3) \notin R_1$ ∴ R_1 is not a diagonal relation

$R_2 = \{(1,1), (2,2), (3,3), (2,3)\}$, here $(2,3) \in R_2$ but $2 \neq 3$ ∴ Not a diagonal relation

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Note:- On any given set A, Exactly "one" diagonal relation is possible

Note: Cardinality of diagonal relation on set A = $|A|$



2 mins Summary

Telegram :
t.me/ vishalRawtiyapw
Capital

- **Topic** Multi-set
- **Topic** Cartesian product
- **Topic** Relation and types of relations

Slide

THANK - YOU